

RESEARCH PROJECT REPORT

Project Title: Explainable AI (XAI) Framework for Ischemic Stroke Localization using Grad-CAM

Domain: Computational Neuroscience / Medical Deep Learning

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1. Abstract

While Deep Learning models achieve high accuracy in medical diagnostics, their "black-box" nature limits clinical adoption. This project presents an interpretable pipeline for Ischemic Stroke detection. By integrating **Grad-CAM (Gradient-weighted Class Activation Mapping)**, the framework provides spatial visual explanations for AI-driven decisions, highlighting necrotic regions in neuro-imaging slices to assist radiologists in rapid diagnosis.

2. Technical Architecture (VGG-Style CNN)

The model utilizes a custom VGG-style Convolutional Neural Network (CNN) optimized for high-dimensional feature extraction from MRI-simulated axial brain slices.

Stage	Layer Type	Output Dimensions	Description
Input	Raw Neuro-Image	$128 \times 128 \times 1$	Grayscale Axial Slices
Block 1	Conv + MaxPool	$64 \times 64 \times 32$	Edge & Contrast Detection

Block 2	Conv + MaxPool	$32 \times 32 \times 64$	Structural Feature Mapping
Block 3	Conv + MaxPool	$16 \times 16 \times 128$	Deep Morphological Features
Block 4	Grad-CAM Target	$8 \times 8 \times 256$	Target ROI Localization
Head	GAP + Dense	1 (Sigmoid)	Binary Classification Score

3. Mathematical Framework (Grad-CAM)

The localization heatmap $L^c_{\text{Grad-CAM}}$ is derived via the following three-step optimization:

- Gradient Computation:** $\frac{\partial y^c}{\partial A^k_{ij}}$
Calculating the gradient of the class score y^c with respect to the feature map activations A^k .
- Global Average Pooling:** $\alpha^c_k = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A^k_{ij}}$
Determining the importance weights α^c_k for each filter.
- ReLU Linear Combination:** $L^c = \text{ReLU}(\sum_k \alpha^c_k A^k)$
Aggregating weighted maps and applying ReLU to suppress non-target pixel contributions.

4. Quantitative XAI Evaluation

The model was validated using a cohort of 100 localized test cases to measure interpretability accuracy:

- Average Pointing Error:** ~12.34 pixels (Centroid-to-Centroid distance).
- Lesion ROI Activation Ratio:** >2.0x (Indicates 2x higher signal density within the infarct zone).

- **Classification Performance:** 98.2% Accuracy on balanced synthetic neuro-datasets.

5. Automated Clinical Report (Sample Output)

Plaintext

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AI NEURO-RADIOLOGY DIAGNOSTIC REPORT

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CASE STATUS: [POSITIVE FOR ISCHEMIC STROKE]
AI CONFIDENCE: 94.82%
LOCALIZATION: High spatial activation in Block 4 (Target Layer)
CLINICAL RISK: URGENT / IMMEDIATE CONSULTATION REQUIRED

Note: Decision supported by Grad-CAM Heatmap Analysis.

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6. Technical Implementation Details

- **Environment:** TensorFlow 2.x & Keras Functional API.
- **Image Processing:** OpenCV & NumPy for alpha-blended clinical overlays.
- **XAI Library:** Custom implementation of [tf.GradientTape](#) for precise backpropagation mapping.